Week 1 Final Report: E-Commerce Data Cleaning and Sales Analysis

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Programme: WeIntern Data Science Internship, Week 1
Dataset: Olist Brazilian E-Commerce Public Dataset (Kaggle)

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1. Title and Objective

The objective of this assignment is to clean an e-commerce dataset, calculate core sales KPIs, identify sales trends and customer behaviour, and communicate the findings with clear visuals. The work follows the Week 1 framework: data cleaning (Task 1), sales analysis (Task 2) and visualization (Task 3).

2. Dataset Overview

The Olist dataset stores its information in separate tables. Five were used:

Table	Contents
Orders	One row per order, with status and timestamps (99,441 rows, 8 columns)
Order Items	One row per item in an order, with product, price and freight (112,650 rows, 7 columns)
Products	Product category and physical attributes (32,951 rows, 9 columns)
Customers	Customer ID, city and state (99,441 rows, 5 columns)
Payments	Payment type, instalments and payment value (103,886 rows, 5 columns)

The handbook's fields were mapped to Olist columns as follows:

Handbook field	Olist column	Note
Order ID	`order_id`	
Customer ID	`customer_id`, `customer_unique_id`	`customer_id` changes with every order; `customer_unique_id` identifies the real customer
Product Name	`product_id`	Olist has no product names
Category	`product_category_name`	Names are in Portuguese
Quantity	rows in Order Items	No quantity column exists, so each item row is counted as one unit
Unit Price	`price`	Item price in BRL, excluding freight; used to calculate revenue

Handbook field	Olist column	Note
Order Date	`order_purchase_timestamp`	Converted to datetime; used for the monthly trend analysis
Payment Mode	`payment_type`	credit_card, boleto, voucher, debit_card (plus 3 rows marked not_defined); an order can have several payment rows
Delivery Status	`order_status`	8 statuses; only delivered orders count as completed sales
City or Region	`customer_city`, `customer_state`	Location of the customer; state used for regional analysis and city for order density

3. Cleaning Summary

The raw files were kept unchanged in `data/raw/`. All cleaning was done on copies, and the cleaned files were saved separately in `data/cleaned/`.

Missing values. Missing values were found in two tables.

Table	Column	Missing	Decision	Reason
Orders	`order_approved_at`	160	Kept	Mostly canceled or unfinished orders
Orders	`order_delivered_carrier_date`	1,783	Kept	Order not yet shipped
Orders	`order_delivered_customer_date`	2,965	Kept	Order not yet delivered
Products	`product_category_name`	610	Filled with `Unknown`	Keeps the records without guessing the category
Products	name length, description length, photos	610 each	Kept	Not used in the analysis
Products	weight and dimensions	2 each	Kept	Too few to matter; not used in the analysis

Order Items, Customers and Payments had no missing values. Missing order dates were explained by order status, so they were treated as normal and not as errors.

Duplicates. No fully duplicated rows were found in any table. Repeated `order_id` values appear in Order Items and Payments, but these are valid: an order can contain several items and can be paid with several payment rows. No rows were removed, because removing them would have deleted real transactions and understated revenue.

Validation.

Check	Result	Action
Zero or negative prices	None found	None needed
Zero freight values	383 item rows	Kept; free shipping is valid
Payments with value 0	9 rows	Kept and noted
Payments with 0 instalments	2 rows	Kept and noted
Date formats	No invalid dates in the 5 date columns	Converted to datetime after validation
Order status values	8 distinct statuses, all lowercase	No change needed
Delivered orders with no delivery date	8	Flagged in `delivered_date_missing_flag`, not deleted
Source spelling in two product columns (`lenght`)	2 columns	Left unchanged to match the source files

Row counts were checked after merging (110,197 rows before and after), which confirmed that no records were duplicated.

4. KPI Summary

A valid sale is an item belonging to an order with the status `delivered`. Revenue is the item price excluding freight, in Brazilian Real (BRL).

KPI	Meaning	Result
Total Revenue	Sum of item prices for delivered orders	13,221,498.11 BRL
Total Orders	Number of unique delivered orders	96,478
Average Order Value	Total Revenue divided by Total Orders	137.04 BRL
Total Units Sold	Order-item records in delivered orders (no quantity column exists)	110,197
Top-Selling Category	Category with the highest sales value	beleza_saude (1,233,131.72 BRL, 9.33% of revenue)
Top Revenue-Generating Product	Product with the highest combined sales value	bb50f2e236e5eea0100680137654686c (63,885.00 BRL)

5. Trend Analysis

Seasonality. Revenue peaked in November 2017 at 987,765.37 BRL, 52.4% above October. Orders rose from 4,478 to 7,289 in the same month. This coincides with Black Friday and suggests a promotion-driven spike, although the data alone cannot prove the cause. Revenue then fell 26.5% in December 2017.

Growth. Monthly revenue grew from 111,798.36 BRL in January 2017 to 838,576.64 BRL in August 2018, an increase of 650.1%. Growth levelled off in 2018, with monthly revenue staying between about 826,000 and 978,000 BRL. For the business, further expansion now depends on new sources of demand.

Category trend. Beleza_saude was the strongest category in 2018, leading four of the last five months. Informatica_acessorios fell from 100,132 BRL in February 2018 to 40,053 BRL in August 2018.

Regional differences. Sao Paulo (SP) generated 38.33% of revenue, and SP, RJ and MG together generated 63.38%. Sales therefore depend heavily on a few states.

Delivery status. Delivered orders make up 97.28% of item revenue. The remaining 2.72% comes from orders not yet delivered or lost; canceled and unavailable orders account for only about 0.71%. The revenue-realisation gap is small.

6. Customer Insights

Repeat customers. Only 3.0% of the 93,358 unique customers ordered more than once, and they generated 5.5% of revenue. Customers with a single order account for 94.49% of revenue, so retention is low.

Segment	Customers	Share of revenue
1 order	90,557	94.49%
2 orders	2,573	4.77%
3 or more orders	228	0.73%

Skew. The top 10% of customers generated 41.1% of revenue, so spending is concentrated in a small high-value group.

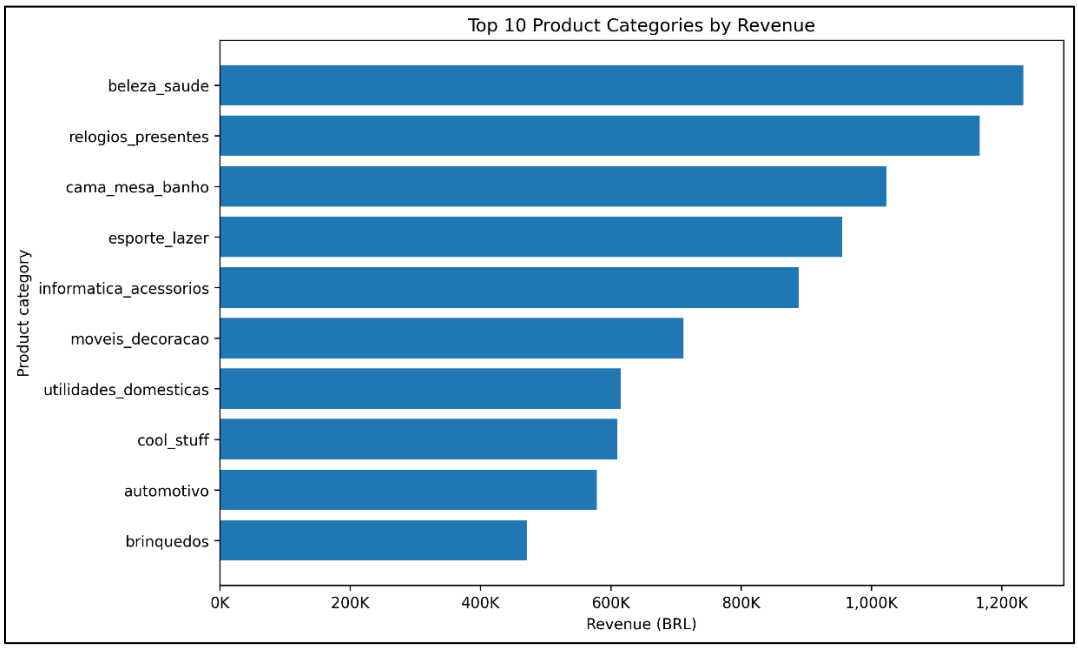
Order density. Sao Paulo had the most orders (15,045, or 15.6% of all delivered orders). The top three cities (Sao Paulo, Rio de Janeiro and Belo Horizonte) together made up 25.2%.

Order value by region. Among states with at least 500 orders, Paraiba (PB) had the highest average order value (217.77 BRL), followed by Para (PA, 184.43 BRL), compared with 137.04 BRL overall.

Category preference. Repeat customers spent more on cama_mesa_banho (13.02% of their spending, against 7.43% for one-time customers), while one-time customers favoured beleza_saude (9.43% against 7.48%).

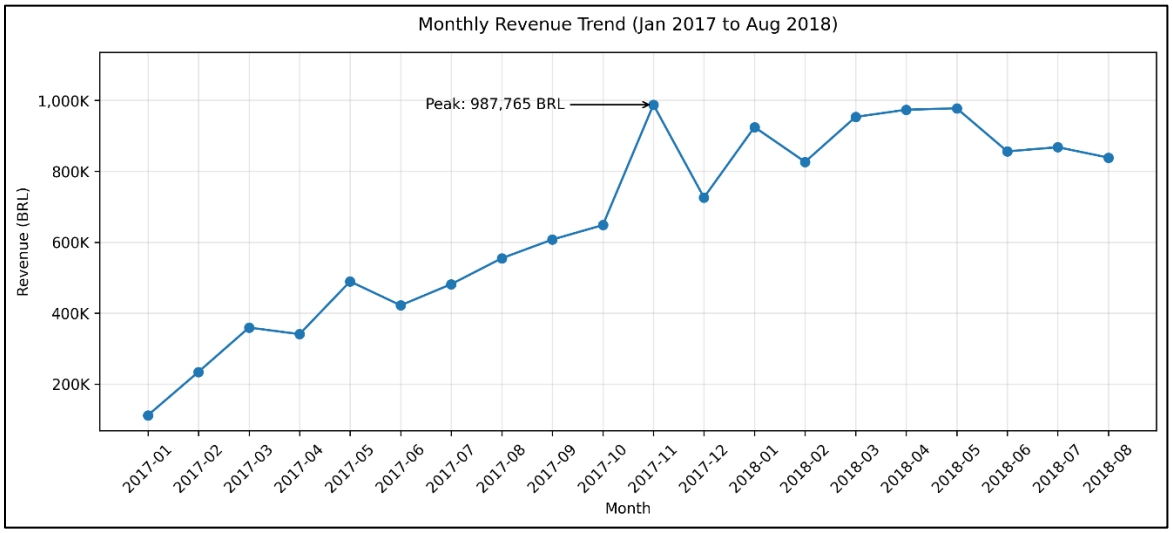
7. Visualization Section

Figure 1: Top 10 Product Categories by Revenue (`revenue_by_category.png`)



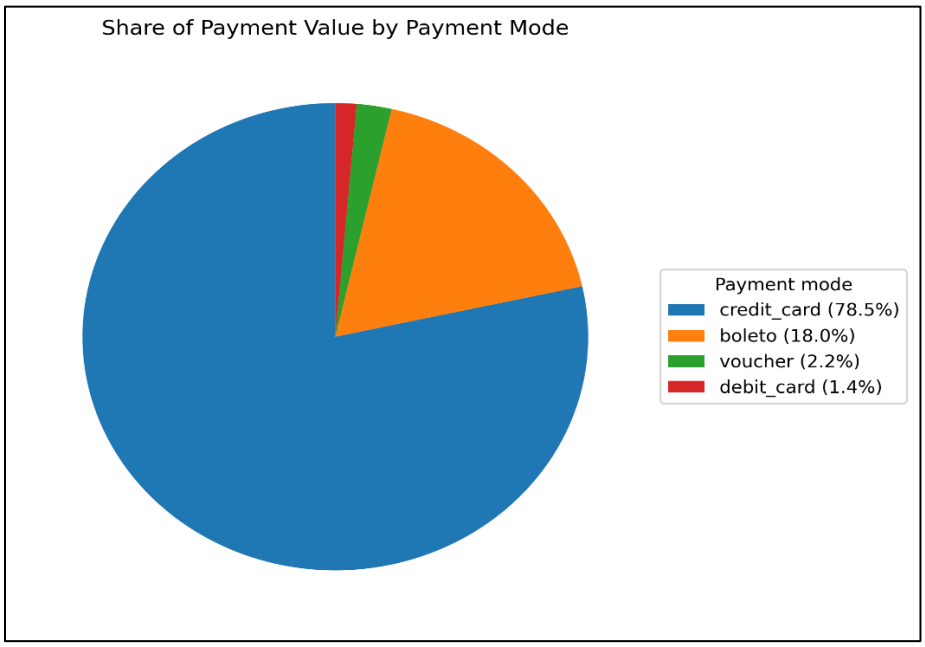
Beleza_saude was the top category with 1,233,131.72 BRL (9.33% of revenue). Revenue is spread across several categories, so no single category dominates sales.

Figure 2: Monthly Revenue Trend (`monthly_revenue_trend.png`)



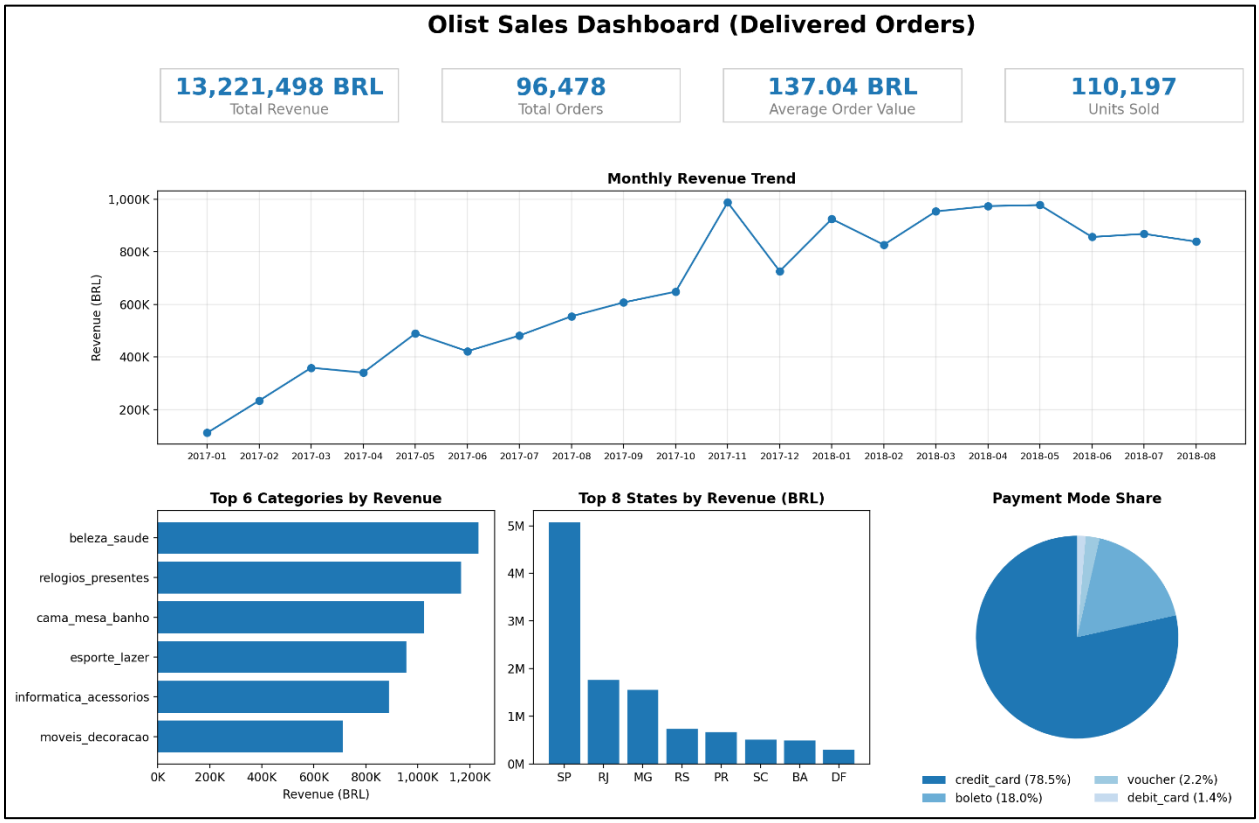
Revenue rose through 2017 and peaked in November 2017 at 987,765 BRL, coinciding with Black Friday. It fell 26.5% in December, then stayed between about 826,000 and 978,000 BRL in 2018, so growth has levelled off.

Figure 3: Share of Payment Value by Payment Mode (`payment_mode_share.png`)



Credit cards made up 78.5% of payment value, followed by boleto (18.0%). Vouchers (2.2%) and debit cards (1.4%) are minor, so the business depends mainly on card payments.

Figure 4: Dashboard Mockup (`dashboard-mockup.png`)



The dashboard combines the main KPIs with the monthly trend, top 6 categories, top 8 states and payment mix. Revenue peaked in November 2017, is spread fairly evenly across the top categories, is concentrated in a few states (SP alone is 38.33%), and is mostly paid by credit card (78.5%).

8. Key Findings

1. Delivered orders generated 13,221,498.11 BRL from 96,478 orders, with an average order value of 137.04 BRL.
2. Beleza_saude was the top category with 1,233,131.72 BRL (9.33% of revenue), so no single category dominates.
3. Revenue peaked in November 2017 at 987,765.37 BRL (+52.4% on October), then fell 26.5% in December.
4. Monthly revenue grew 650.1% between January 2017 and August 2018, but stayed within about 826,000 to 978,000 BRL throughout 2018.
5. SP, RJ and MG together generated 63.38% of revenue, so sales depend heavily on a few states.
6. Only 3.0% of customers ordered more than once, and they generated just 5.5% of revenue, so retention is low.
7. The top 10% of customers generated 41.1% of revenue.
8. Credit cards accounted for 78.5% of payment value, followed by boleto (18.0%).

9. Recommendations

These are suggestions based on the findings above, not findings themselves.

- **Improve retention.** Run follow-up offers or loyalty rewards, because about 97% of customers buy only once.
- **Prepare for November.** Plan stock and promotions ahead of the seasonal peak.
- **Reduce regional dependence.** Target growth outside SP, RJ and MG. States such as PB and PA have the highest average order values.
- **Protect high-value customers.** Offer priority service or exclusive offers to the top 10% of customers, who generate 41.1% of revenue.

10. Limitations and Assumptions

- Revenue counts delivered orders only and excludes freight. Amounts are in Brazilian Real (BRL).
- Units sold are counted as order-item records, because the dataset has no quantity column.
- Product IDs are used because product names are not available, and category names are in Portuguese.
- Data for late 2016 and from September 2018 is incomplete, so trend conclusions use January 2017 to August 2018.
- Payment shares are based on payment value, which includes freight, so they do not match item revenue exactly.
- Repeat customers are identified with `customer_unique_id`. The findings show association, not proof of cause.